PATIENT SESSION

LIVE

M. Alvarez
F · 34y • Routine Screen

Session ID	SS-2026-0142
Method	Landolt C · YOLOv8
Eye	OD (Right)
Started	14:32:08

TEST CONTROLS

- Calibrate Screen
- Enable Privacy Mode
- Pause Session
- Reset System

TIME REMAINING

15 s

Attempts 0/15

COGNITIVE METRICS

Response Time	0.0 s	↓0.3s
Hesitation	None detected	Normal
Ambient Lighting	329 lux	OK
Head Stability	98.5%	Stable

LIVE · 1080p | WebRTC | 14:41:12

Awaiting video feed

GET /video_feed · waiting on Flask backend

ESTIMATED DISTANCE
60 cm
40 50 70 90 cm

Status TRACKING ACTIVE FPS 31.8 Inference 11 ms Confidence 98.4%

GPU RTX A4000 · 42°C

CLINICAL OUTPUT

• RECORDING

DISTANCE	SCORE
60 cm	0 /15
Within range	80% accuracy

VISUAL ACUITY · SNELLEN

20/40

Live estimate · Confidence 92%

VISION CHART · LANDOLT C

Size · 10.0'

C

↑ Up ↓ Down ← Left → Right

Patient gestures the direction of the **opening**.
Awaiting input…

neural_processing.log

STREAMING

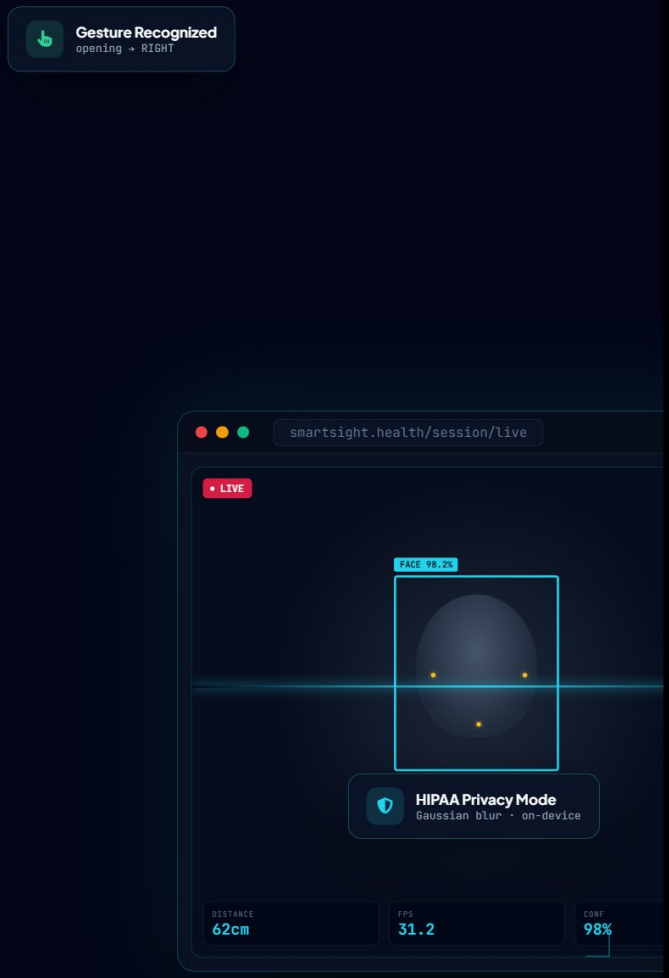
```
14:32:21 [TGT]  New target → opening RIGHT @ 20/30
14:41:00 [SYS]  Dashboard ready · awaiting clinician input
14:41:00 [WARN] /test_status unreachable · running local simulator
14:41:06 [CMD]  Assessment started · 15 trials configured
```

SmartSight
v2.1 · CLINICAL

Vision Testing, Reinvented by AI.

An AI-powered telehealth platform that turns any webcam into a **clinical-grade vision station** — automating Snellen acuity with computer vision.

YOLOv8 | OpenCV | Flask | HTML5 · JS



02 · THE CLINICAL PROBLEM

The telehealth vision gap

A test designed for a controlled clinic breaks the moment it leaves the room.

LEGACY · SINCE 1862

The wall chart works — only in the clinic

The Snellen chart relies on a controlled room: a fixed 20-foot distance, calibrated lighting, and a trained observer watching every response.



REMOTE · THE REALITY

Every control disappears at home

No distance control
Acuity is meaningless if you don't know how far the patient sits from the screen.

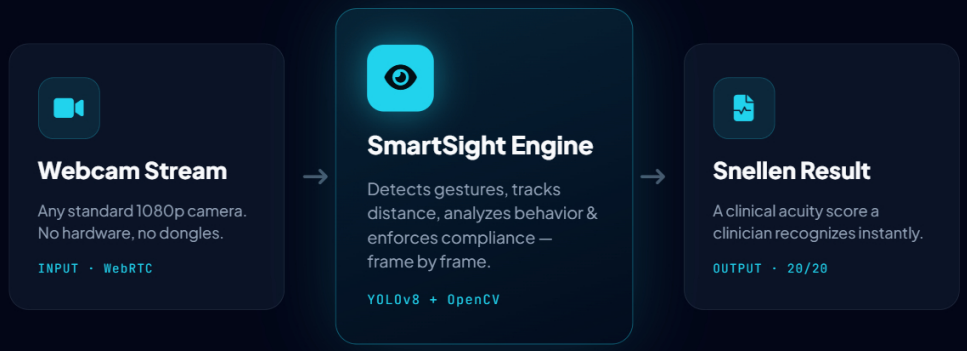
No cognitive analysis
A 3-second hesitation is a clinical signal — invisible over a plain video call.

No integrity guarantee
Nothing stops a patient from leaning in, squinting, or reading from off-screen.

03 · THE SMARTSIGHT SOLUTION

One layer between webcam and diagnosis

SmartSight bridges real-time computer vision with clinical acuity scoring — entirely in the browser.



Zero-friction
Runs on the device the patient already owns — a browser and a webcam.

Clinically rigorous
Distance, lighting and behavior are verified before any score is trusted.

Private by design
On-device face blur keeps protected health data off the wire.

04 · CORE ARCHITECTURE

Built for real-time inference

A latency-aware pipeline with a majority-voting noise filter at its core.

smartsight/pipeline.py

```python
# 1 · Downsample frame → cut inference latency
frame = cv2.resize(raw, (640, 480))

# 2 · YOLOv8 gesture detection
pred = model(frame, conf=0.55)
gesture = decode_direction(pred)

# 3 · Majority vote over rolling window
window.append(gesture)  # maxlen=15
if len(window) ⩾ 10:
    stable = majority_vote(window)
    if stable.conf > 0.7:
        register_answer(stable)
```

01 Capture & Downsample
1080p → 640×480 before inference

02 Detect & Vote
10/15-frame majority filter rejects flicker

03 Map to Acuity
Stable answer → Snellen score

YOLOv8
Gesture detection

OpenCV
Face & distance

Flask
Inference server

HTML5 · JS
Clinician UI

Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Markup and share

05 · INTELLIGENT CLINICAL FEATURES

Four checks that earn clinical trust

Each algorithm maps to a real measurement-integrity or compliance requirement.

01

Distance Estimation

OpenCV face geometry holds the patient inside the calibrated test zone — the prerequisite for any valid acuity reading.

50–70 cm tracking zone

02

Cognitive Delay Tracking

Response latency is timed per trial. Slow answers are flagged as hesitation — a signal of uncertainty, not just error.

Flag at > 2.5 s

03

Anti-Cheat Gaze Detection

If the face leaves frame or looks away, the trial auto-pauses — closing the door on off-screen prompting.

Auto-pause on face loss

04

HIPAA Privacy Mode

A live Gaussian blur masks the face before any frame is rendered or stored — protected data never leaves the device unmasked.

On-device face blur

06 · SIMULATED LIVE DEMO

A trial, end to end

An adaptive loop that shrinks the optotype with every correct answer — homing in on the true threshold.

STEP 01

Calibrate

Face locked, distance & lighting verified. The test unlocks only when conditions are valid.



STEP 02

Present

A Landolt C appears. The patient gestures the direction of the gap — no keyboard needed.



STEP 03

Adapt

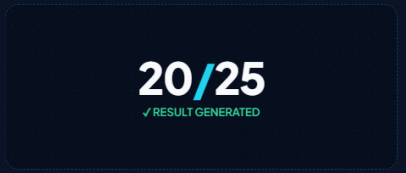
Each correct match shrinks the optotype to the next acuity level, narrowing toward the limit.



STEP 04

Score

The smallest reliably-read size converts to a Snellen acuity — the clinical result.



07 · IMPACT & FUTURE SCALE

Democratizing digital eye care

From a clinic-bound test to a screening anyone can take from home.

0 — **Specialized hardware**
A webcam and a browser replace the chart, the room, and the observer.

~3 min — **Per full assessment**
Calibration to Snellen result in a single guided session.

31 fps — **Real-time inference**
Smooth, responsive tracking on commodity hardware — no GPU farm.

The Road Ahead

Native Mobile
Bring the test to the phone every patient already carries.

WebRTC Streaming
Production-grade, low-latency video at clinical scale.

EHR Integration
Results flow straight into the records clinicians already use.

The hardest part of eye care has never been the chart — it's access.